VIJAY BOLLE

AI Engineer | Enterprise Agentic AI Systems | Healthcare
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PROFESSIONAL PROFILE

Experienced in architecting LLM-powered applications, RAG pipelines, and multi-agent AI systems using LangChain and LangGraph to automate complex healthcare prior authorization and claims processing workflows. Skilled in integrating intelligent AI agents across multiple layers of enterprise production systems to minimize human intervention, streamline operations, and improve efficiency. Hands-on expertise in developing multi-stage OCR pipelines using Azure AI Document Intelligence with Claude Vision-powered fallback mechanisms, achieving 89.4% reviewer acceptance and identifying 42% potential cost savings without compromising accuracy. Proficient in Python, FastAPI, REST APIs, vector databases, prompt engineering, LLM evaluation, and production monitoring, with a strong focus on delivering secure, scalable, and reliable AI solutions in HIPAA-regulated healthcare environments.

WORK EXPERIENCE

GENZEON, Exton, PA — 06/2025 – Present

AI Engineer

Engineer on **WISeR**, Genzeon's Medicare prior-authorization platform: **document intelligence**, **validations** and **agentic decisioning** under **CMS compliance** on Azure Government. Own the LLM pieces of the document pipeline, the measurement of how they perform on production data, and the safe rollout of AI agents.

- ✓ Designed and deployed a production-grade, LLM-powered OCR pipeline using Azure AI Document Intelligence and Claude Sonnet to intelligently identify coversheets within multi-page healthcare documents and accurately extract critical fields for Medicare prior authorization workflows, achieving 99% field extraction accuracy. Implemented confidence-based field reprocessing, Claude Vision-powered fallback mechanisms, deterministic field validation, hallucination safeguards, and field-level provenance tracking to ensure reliable, scalable, and auditable document processing.
- ✓ Contributed to the design, development, and production enhancement of WISeR, Genzeon's enterprise-scale Medicare prior authorization and claims processing platform hosted on Azure Government. Developed and integrated automated healthcare workflows for multi-channel request intake (Portal, Fax, and esMD), intelligent document processing, eligibility verification (HETS), provider enrollment validation (PECOS), program integrity screening (UPIC), and CMS-compliant decision routing. Supported seamless integration with Clinical Operations systems for human-in-the-loop review, automated decision communication.
- ✓ Enhanced and optimized a production-grade, LangGraph-based multi-agent system for automated Medicare prior authorization screening using Python, FastAPI, Azure OpenAI (GPT-4o-mini), and Chroma RAG. Integrated parallel validation workflows for patient eligibility (HETS), provider enrollment (PECOS/NPPES), program integrity (UPIC), duplicate detection, and CPT/ICD-10 code validation. Designed and implemented end-to-end UPIC screening with dynamic backend integrations, dual-NPI verification, fail-open error handling, and deterministic decision guardrails to ensure accurate, reliable, and CMS-compliant dismissal decisions while reducing manual intervention.
- ✓ Enhanced a production-grade, LangGraph-based Medicare prior authorization dismissal agent by migrating critical CMS validation rules into a unified, dynamically configurable rule engine, ensuring consistency between V1 and V2 agent workflows. Designed and implemented a production-safe shadow deployment framework with isolated agent execution, PostgreSQL-based result tracking, automated V1/V2 decision comparisons, REST APIs, and monitoring dashboards to evaluate agent accuracy, maintain rule parity, and support controlled rollout without impacting existing production decisions.
- ✓ Conducted extensive performance evaluations and comparative analysis of V1 and V2 prior authorization screening agents over several months, analyzing results from 4,000 prior authorization requests across development and testing environments. Identified decision inconsistencies, validation gaps, and optimization opportunities through continuous accuracy assessments and workflow analysis. Implemented targeted enhancements and supported the deployment of the most reliable agent version into production, reducing manual intervention and dependency on the Clinical Operations team while improving screening consistency and operational efficiency.
- ✓ Evaluated AI output against 28,038 reviewer corrections across 4,609 production coversheets; the Claude fallback reached 89.4% reviewer acceptance and the study identified 42% avoidable model-call cost.

- ✓ Designed and enhanced automated Medicare claims processing workflows using Python, FastAPI, and LangGraph-based state machines to streamline claims screening, eligibility validation, and CMS-compliant decision routing. Implemented UPIC program-integrity screening, automated Not Applicable (N/A) classification for Part B prepayment claims, non-response eligibility rules, and human-in-the-loop review workflows. Integrated esMD Additional Documentation Request (ADR) response mapping using unique tracking numbers, enabling accurate claims-to-document association, reduced manual intervention, and reliable end-to-end claims processing.
- ✓ Implemented Operator, Clinical and Viewer roles with Microsoft Entra, read-only enforcement across 15 API routers, PHI hidden from read-only users, and JWT validation on the API.
- ✓ Designed, configured, and deployed end-to-end Development, UAT, and Production environments from scratch on on-premises Enterprise Linux servers. Managed system configuration, RPM packaging, service installation, and deployment of AI-powered applications, integrating databases, LLM services, and cloud-hosted models for OCR and reasoning workflows. Troubleshot environment-specific infrastructure, dependency, compatibility, and computational resource challenges to ensure reliable service execution, seamless system integration, and consistent application performance across environments.
- ✓ Designed and developed domain-specific ontologies and semantic knowledge layers to enhance the accuracy, reliability, and contextual reasoning of frontier LLMs and on-premises AI models within healthcare applications. Integrated authoritative Medicare and Medicaid knowledge resources to ground model responses, minimize hallucinations, and improve regulatory alignment. Implemented knowledge-driven validation and evaluation frameworks to assess model accuracy, consistency, and reasoning quality, enabling more reliable and explainable AI-driven decision support in production healthcare workflows.
- ✓ Performed end-to-end troubleshooting and root cause analysis across application, service, and infrastructure layers in enterprise production environments. Diagnosed complex, business-critical issues through application logs, service monitoring, Linux server diagnostics, and system-level investigations, identifying failures across APIs, databases, dependencies, and infrastructure. Prioritized incidents based on severity and business impact, implemented targeted fixes, and improved system stability, service availability, and operational reliability.
- ✓ Evaluated open-source LLMs, including Qwen and Llama, in on-premises Linux environments, benchmarking reasoning capabilities, response accuracy, inference latency, computational efficiency, and hallucination risks against frontier models to identify reliable and cost-effective solutions for enterprise healthcare AI applications.

ED & IMMIGO, Hyderabad, India 05/2020 – 07/2022

Web Application Developer

Delivered about 10 production web applications for a study-abroad consultancy, from React and Angular front ends to Python REST back ends on Linux servers.

- ✓ Delivered about 10 production web applications end to end for a study-abroad consultancy, including student portals, educational and services portals, and marketing sites.
- ✓ Built front ends in React, Angular, and HTML/CSS/JavaScript, then integrated them with Python back ends in Django, Flask, and FastAPI through REST APIs backed by PostgreSQL.
- ✓ Deployed and maintained the applications on enterprise Linux servers behind Nginx, debugging and resolving back-end and integration issues in production.
- ✓ Gathered requirements directly from clients, handled post-launch fixes and change requests, and managed source control for each application with Git.

TECHNICAL SKILLS

Agentic AI & Orchestration	LangGraph (StateGraph, supervisor node, rule packs), LangChain, ReAct supervisor agents, deterministic tool design, guardrails and anti-hallucination checks, human-in-the-loop approval, confidence-gated fallbacks, MCP servers (FastMCP)
Models & AI Platforms	Anthropic Claude (Python SDK, vision input, tool-use structured output, structured JSON output, stop-reason handling, token budgets), Azure OpenAI (GPT-4o-mini, Government cloud), Azure AI Foundry, OpenAI-compatible APIs, open-source LLM evaluation (Qwen, Llama) on on-prem vLLM / llama.cpp, MedGemma, sentence-transformers, prompt engineering, model benchmarking (accuracy, latency, cost, hallucination risk)
Document AI	Azure AI Document Intelligence custom models, OCR pipelines, PyMuPDF, pypdf, Pillow, page rendering and deskew, docling, onnxtr / docTR, YOLO layout detection, OpenCV, scispaCy clinical NER

Context, Semantics & Data	RAG and vector search (Chroma, pgvector, FAISS, sentence-transformer embeddings), domain ontologies and semantic knowledge layers, knowledge graphs (Apache AGE), Medicare and Medicaid knowledge resources, PostgreSQL (PL/pgSQL functions, SQLAlchemy 2.0, raw-SQL migrations with rollback scripts, postgres_fdw), Redis, governed read-only data access, read-only production investigation, snapshot and backfill scripts
Evaluation, Security & Observability	AI evaluation against reviewer corrections (error taxonomies, cost-waste analysis), golden-set evaluation gates, regression suites, prompt-injection tests, Microsoft Entra ID / Azure AD (app roles, JWT/JWKS, MSAL), RBAC and least privilege, Key Vault, Managed Identity, PHI controls, Azure Monitor (Application Insights, Log Analytics), production service monitoring, log analysis and server diagnostics, cross-layer root cause analysis, severity-based incident triage
Engineering & Runtime	Python 3.11 / 3.12, FastAPI, Pydantic v2, asyncio, httpx, structlog, tenacity, pytest, Django, Flask, REST APIs, gunicorn / uvicorn, React 18, Angular, TypeScript, HTML / CSS / JavaScript, responsive web design, Vite, Tailwind, axios, SSE streaming, Chart.js, Vitest and Testing Library, WCAG 2.2 AA / Section 508; Azure (US Government cloud, App Service, Functions with timer triggers, Blob Storage, Azure Database for PostgreSQL, Container Registry, Azure Artifacts, Microsoft Graph), Docker / Podman, Git, Azure DevOps, Azure Pipelines, GitHub Actions, CI/CD, on-premises Enterprise Linux (Dev / UAT / Prod environment provisioning, RPM packaging, service installation and configuration), Nginx, dependency and compatibility troubleshooting
Healthcare	CMS WISeR prior authorization (Operational Guide), Medicare claims processing (Part B prepayment review), HETS eligibility, PECOS, NPPES, UPIC, esMD, UTN, gold card, ADR (Part A / Part B), ICD-10 / CPT / HCPCS, NPI / MBI, FHIR R4 / R4B, HIPAA compliance and PHI handling, 42 CFR 422.2267 letter rules

EDUCATION

Master of Science, Computer Technology 08/2023 – 12/2024
EASTERN ILLINOIS UNIVERSITY, Charleston, IL · GPA 3.7

CERTIFICATIONS

- ✓ Microsoft Foundry partnership hackathon – shipped and deployed agents to AI Foundry
- ✓ Claude Certified Architect – Foundations
- ✓ Microsoft Certified: Azure Fundamentals